

SOFTWARE ENGINEERING 2

FINAL PROJECT PROPOSAL

Theme: AI-Integrated Information System for Real-World Problems

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Course / Year: BSCS-4

Subject: Software Engineering 2

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Date: August 7, 2026

PROJECT TITLE

AI-Enhanced Campus Lost and Found Management System

Live System: <https://lost-and-found-system-7ro8.onrender.com/>

I. Background of the Study

The Campus Lost and Found Management System started as a web-based platform built to replace the paper logbooks and bulletin-board postings that educational institutions traditionally rely on for handling lost and found items. The system is fully developed and deployed live on Render. It enables students and faculty to register accounts, report lost or found items complete with photos and descriptive details, search and filter item listings, submit formal claim requests with proof of ownership, and track claim status. Dedicated administrative dashboards allow staff to review reports, process claims, manage categories, and configure campus locations.

While the core system included a baseline item-matching feature, it operated strictly via a rule-based scoring algorithm. This algorithm compared reports using categorical tags, location text, date proximity, item names, and keyword overlaps, producing a similarity score out of 100. Although effective for carefully written reports, this approach revealed a significant limitation: it never analyzed the actual photos uploaded by reporters, despite photos being present in nearly every report. Furthermore, it could not parse natural language queries, forcing students to guess exact keywords rather than describing lost items in plain language.

As item reports accumulate, the discrepancy between what the algorithm evaluates (typed text) and what is visually available (item photos, descriptive nuance) leads to missed reunifications. By integrating Artificial Intelligence (AI) — specifically vector image similarity and multimodal visual analysis — the system can directly compare item photos and understand natural language inquiries, closing the accuracy gap without increasing the manual workload of administrative staff.

The proposed enhancement improves match accuracy and user accessibility while maintaining the existing platform's operational workflow, administrative approval safeguards, and underlying database structure.

II. Problem Statement

The proposed enhancement directly addresses the following operational problems:

1. **Unused Visual Data in Matching:** The legacy matching algorithm relied exclusively on typed text fields (category, location, name, description) and never evaluated uploaded photos, causing visually identical items described with different phrasing to be missed.
2. **Rigid Keyword Search:** Students were required to know exact search terms to locate items; the system lacked natural language processing capabilities to handle plain-language inquiries.
3. **Vulnerability to Inconsistent Descriptions:** Matching accuracy depended heavily on user phrasing. Vague or inconsistent descriptions (e.g., "black bag" vs. "charcoal backpack") caused rule-based scoring to fail.
4. **Lack of Explainable AI Match Reasoning:** Staff reviewing candidate matches were presented only with numerical scores without visual confirmation or human-readable explanations detailing why two items were flagged.

III. Proposed System

The AI-Enhanced Campus Lost and Found Management System is a production-grade web platform designed to streamline lost and found operations across university campuses. The platform enables students and faculty members to log lost or found items with images, filter active listings, submit claims supported by proof of ownership, and track claim statuses in real time.

Campus administrators utilize a dedicated control panel to inspect incoming reports, approve or reject claim requests, manage category and location directories, and monitor overall campus reunification metrics.

The platform embeds an AI Layer powered by modern visual embeddings and multimodal Generative AI (Gemini API & CLIP feature extraction). This layer compares item photos for visual similarity, provides plain-language match explanations, powers a 24/7 conversational assistant for natural-language search, and processes direct photo uploads within the chat widget to identify lost items automatically.

IV. AI Integration

AI Feature: Hybrid Visual Matching Engine & Multimodal Conversational Assistant

The system integrates visual embedding algorithms alongside the Gemini API to extend rule-based scoring with real computer vision and conversational intelligence.

Key Capabilities of the AI Feature:

- **Image Visual Similarity Scoring:** Computes vector cosine similarity between uploaded item photos to generate an accurate visual match score merged with the rule-based score.
- **Plain-Language Match Explanations:** Generates human-readable reasoning explaining why two items were identified as candidate matches.
- **Conversational AI Chat Assistant:** Answers student questions regarding lost or found items via a 24/7 chat widget.
- **Natural-Language Entity Extraction:** Automatically parses typed messages to extract item categories, colors, brands, and key terms for database searching.
- **Multimodal Chat Photo Analysis:** Accepts image uploads directly inside the chat interface, automatically identifying item categories, colors, brands, and distinguishing attributes to find database matches.

V. Target Users

Primary Users

- **Students:** Report lost/found items, search listings via text or photo, interact with the AI assistant, submit claims, and track claim status.
- **Faculty / Campus Staff:** Report items found within departments, track reported items, and guide students through reunification.

Secondary Users

- **Campus Security / Admin:** Review pending reports, inspect AI match scores, evaluate ownership claims, manage campus locations/categories, and generate system statistics.
- **System Administrator:** Oversee system health, user accounts, security policies, and technical maintenance.

VI. Project Objectives

General Objective

To enhance the existing Campus Lost and Found Management System with AI-powered visual matching and a multimodal conversational assistant to improve match accuracy and search accessibility for students and university staff.

Specific Objectives

1. To integrate AI computer vision and feature vector comparison into the existing item-matching pipeline.
2. To deploy a conversational AI assistant supporting natural-language text queries and direct photo uploads.

3. To improve lost and found item reunification rates beyond text keyword matching alone.
4. To provide staff with plain-language, explainable match justifications for AI-flagged items.
5. To preserve established administrative approval, security, and claim verification workflows.

VII. Scope

The scope of the enhancement includes:

- Automated AI-powered visual similarity scoring between item photos.
- A multimodal chat widget supporting natural-language inquiries and image upload queries.
- Plain-language explanation generation for high-confidence visual matches.
- Seamless integration with existing report creation, claim submission, user authentication, and admin review workflows.

VIII. Limitations

- **Human Verification Required:** The system does not automate physical item handover; administrative staff must verify physical proof of ownership before releasing an item.
- **Advisory AI Scoring:** AI-generated visual match scores act as candidate suggestions and do not trigger automatic claim approvals.
- **Domain-Specific Chatbot:** The AI assistant is constrained exclusively to lost-and-found queries and will not provide legal, medical, academic, or financial advice.
- **Hardware Scope:** The system does not integrate with CCTV camera networks or IoT physical location tags.

IX. Proposed System Features

1. User Authentication & Role-Based Access: Secure session management for Students, Faculty, and Administrators.
2. Item Reporting with Photo Upload: Complete reporting forms with image uploads and custom location tagging.
3. Dynamic Search & Filtering: Filter items by category, location, status (lost/found/claimed), and date range.
4. Rule-Based Matching Engine: Initial candidate filtering based on category, location, date, and text overlap.
5. AI Visual Image Comparison: Feature vector extraction (CLIP embeddings) and cosine similarity visual scoring.

6. **Conversational AI Assistant:** Embedded chat widget supporting plain-language queries and item assistance.
7. **In-Chat Photo Search:** Direct photo attachment within the chat widget for automated visual identification and matching.
8. **Claim Request Management:** Submission of ownership proof and real-time status tracking for claimants.
9. **Admin Dashboard & Analytics:** Comprehensive management of pending reports, claim verification, location approvals, and CSV exports.
10. **Automated Notification System:** Email notifications for report approvals, potential item matches, and claim updates.

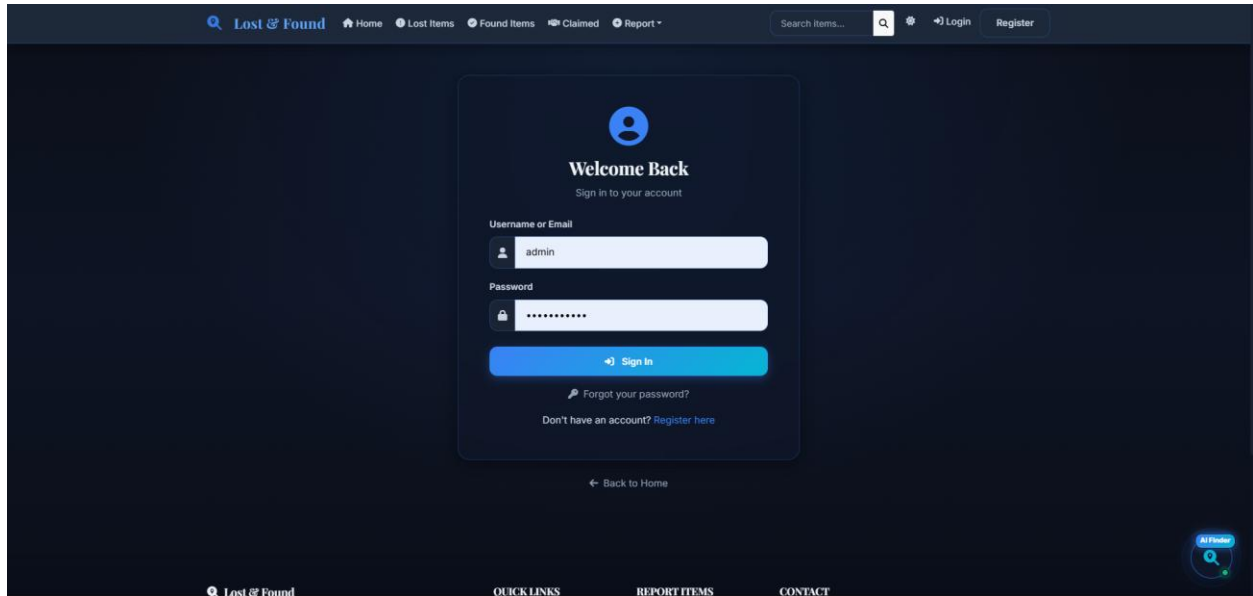
X. Expected Benefits

- **Higher Reunification Rates:** Significantly increases match discovery by evaluating visual photo evidence.
- **Reduced Staff Workload:** Streamlines the review process by highlighting strong candidate matches automatically.
- **Self-Service Accessibility:** Allows students to search intuitively using plain language or item photos.
- **Explainable AI Decision Making:** Gives staff clear, transparent reasoning for flagged matches.
- **Faster Item Recovery:** Decreases the average time items remain unclaimed in campus holding offices.

XI. Initial User Interface

Screen 1: Login & Authentication Page

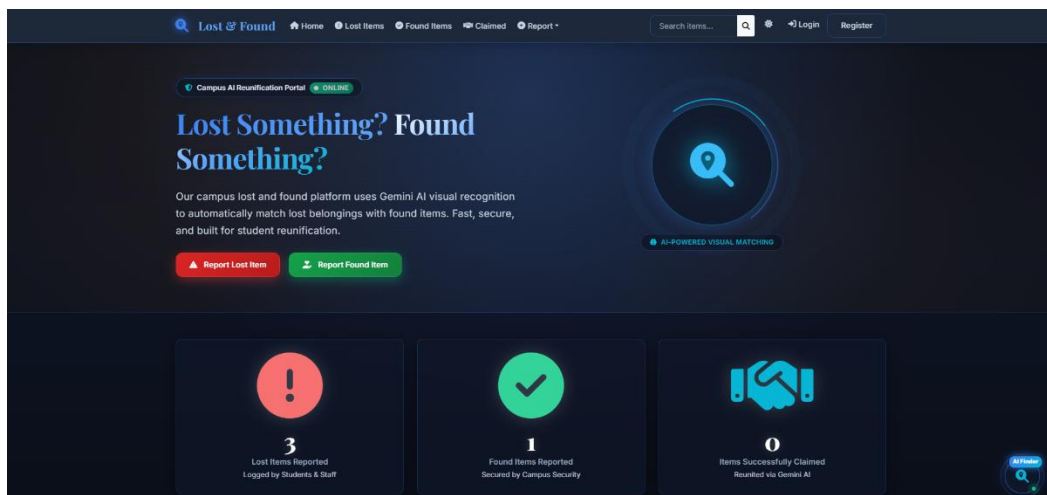
Clean, responsive login portal providing role-aware access for students, faculty, and campus administrators.



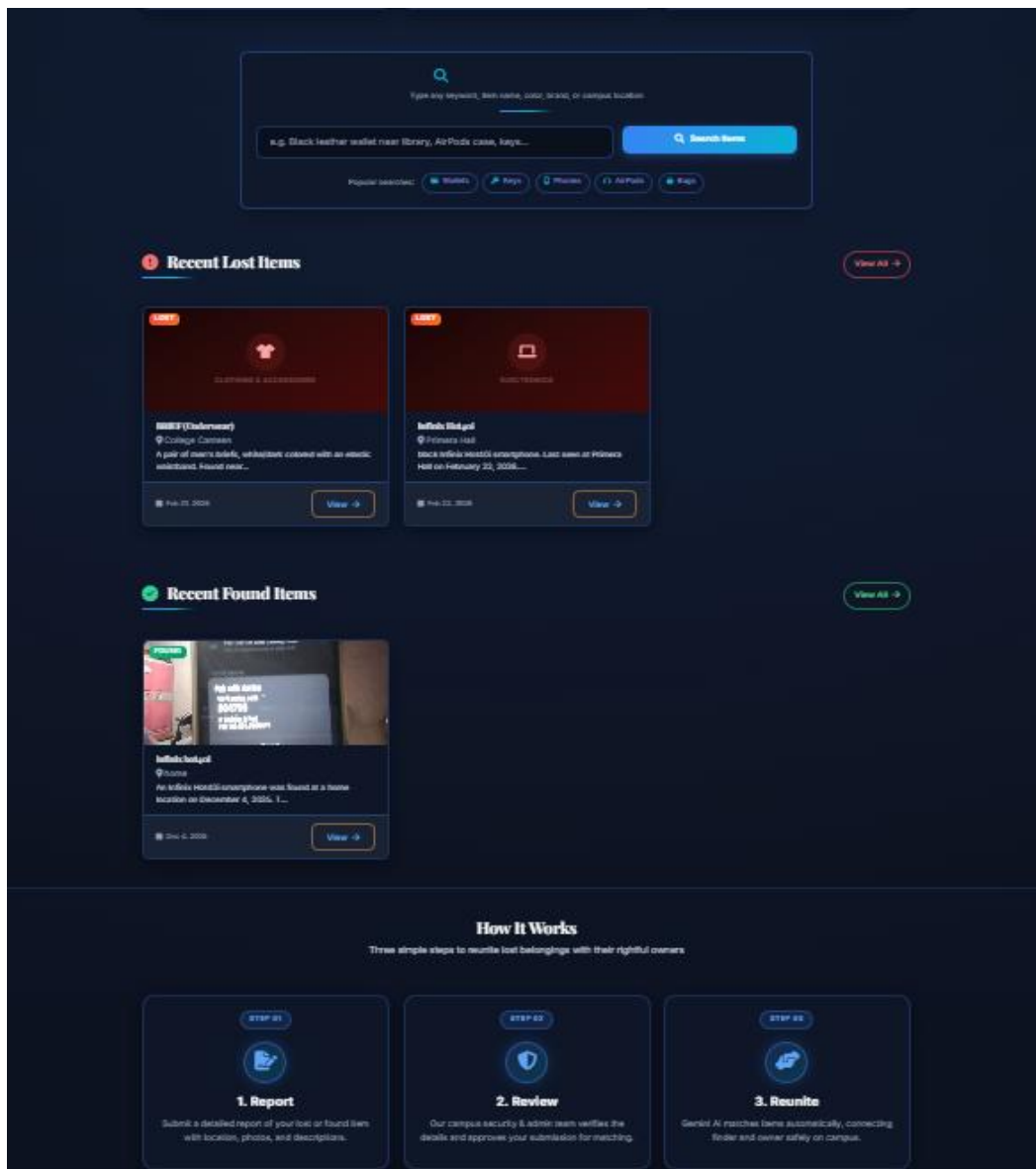
Login page — role-aware sign-in for students, faculty, and admins.

Screen 2: Homepage & Report Item Interface

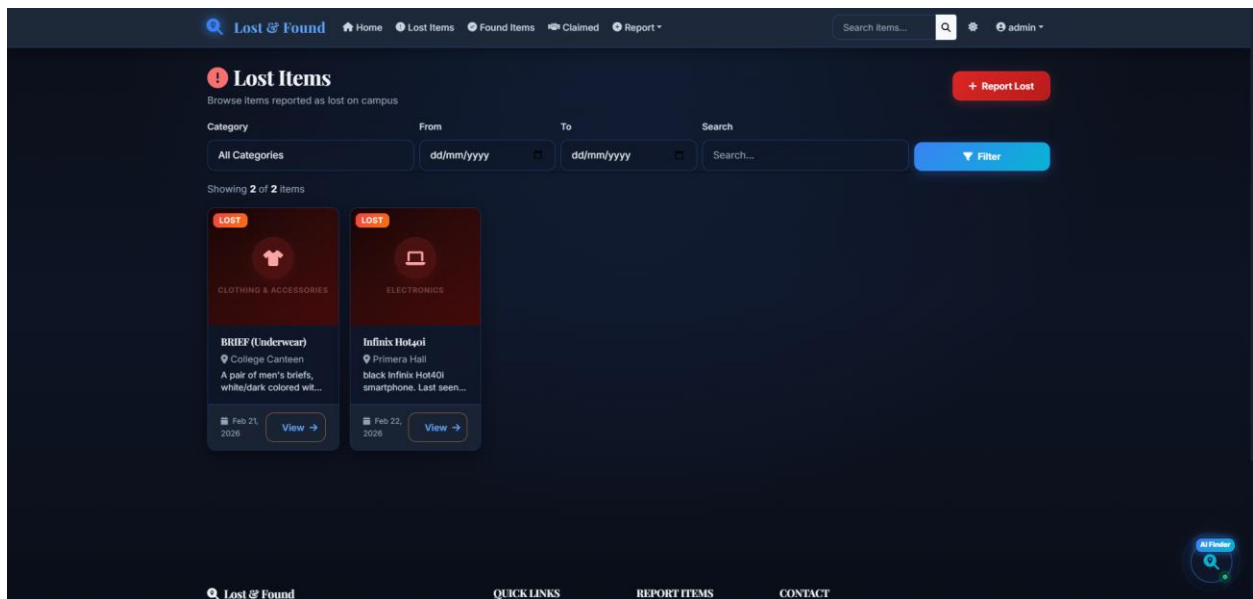
Modern UI featuring active item stats, report forms with file dropzones for photo uploads, and active listing grids.



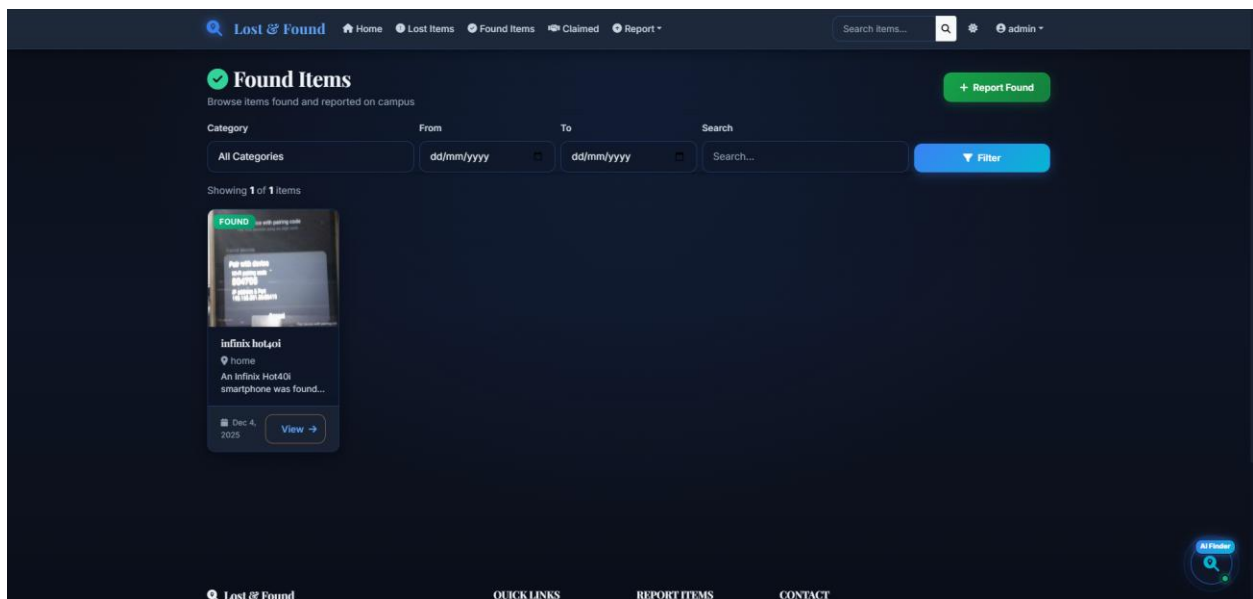
Homepage hero — live reunification stats and quick report actions.



Homepage — recent lost and found item listings.



Lost Items page — category, date range, and keyword filtering.



Found Items page — browsing items reported found on campus.

Report Lost Item form — item details, photo upload, and contact info.

Lost & Found

[Home](#)[Lost Items](#)[Found Items](#)[Claimed](#)[Report](#)

Search items... admin

[Home](#) / [Report Found Item](#)

 **Report Found Item**

Thank you for helping! Fill out the form below to report a found item. Your report will be reviewed by an admin before being published.

Item Details

Item Name *

e.g., Blue Backpack, iPhone 13, etc.

Category *

Select Category

Description *

Describe the item in detail (color, brand, size, distinguishing features, etc.)

Max 1000 characters0/1000

Location Found *

Select Location

Date Found *

dd/mm/yyyy

Select from list or choose "Other" to type your own

Item Image (Optional)

Choose File

No file chosen

Accepted formats: JPG, PNG, GIF, WebP. Max size: 5MB

Contact Information

Your Name *

admin

Your Email *

rbwork2002@gmail.com

Additional Contact Info (Optional)

Phone number or alternative contact method

☐ I confirm that the information provided is accurate and I understand that my report will be reviewed before publishing.

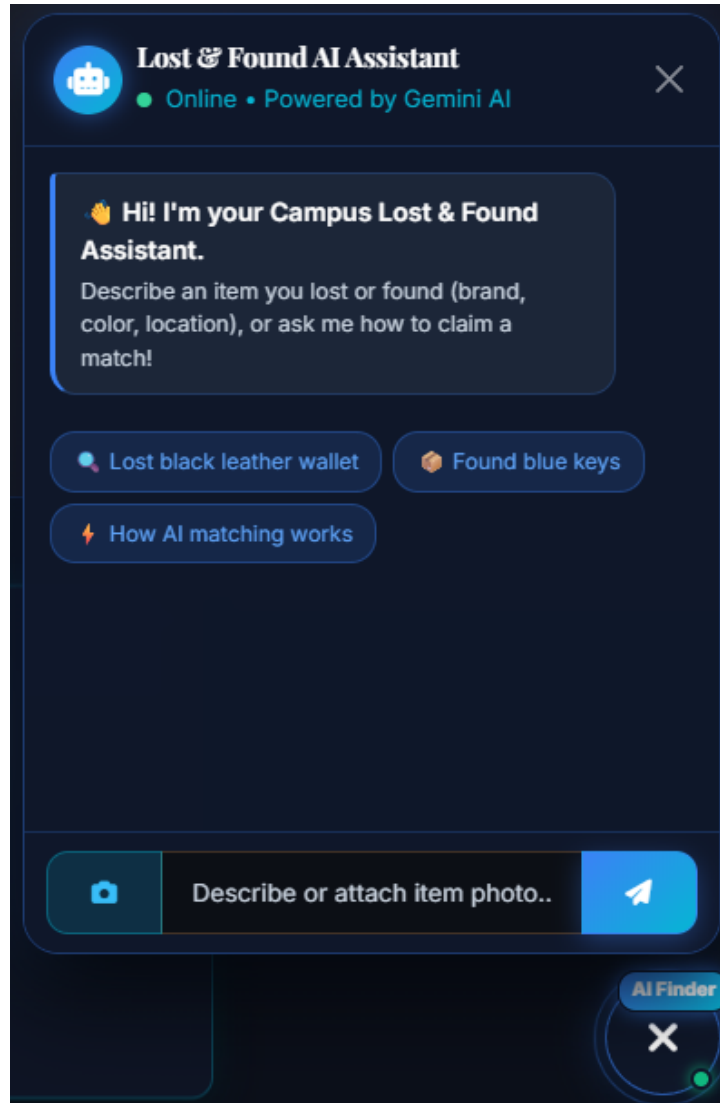
 **Submit Report**

 **Go Back**

Report Found Item form — mirrors the lost-item reporting flow.

Screen 3: AI Chat Assistant & Photo Search Panel

Interactive floating chat widget (bottom-right “AI Finder”) supporting text input, camera file attachments, direct item match cards, and direct navigation links (/items/:id).



AI chat widget with photo attachment, quick-reply prompts, and Gemini-powered natural language search..

XII. Technology Stack

Frontend: EJS (Embedded JavaScript Templates), Bootstrap 5, Vanilla CSS3, FontAwesome

Backend: Node.js, Express.js

Database: MongoDB Atlas (Mongoose ODM with custom compound indexes)

AI Integration: Gemini API (gemini-2.0-flash / gemini-2.5-flash) & Hugging Face Inference API (openai/clip-vit-base-patch32 embeddings)

Storage & Hosting: Cloudinary (Image Management) & Render.com (PaaS Web Service Deployment)